Machine Learning Versus Deep Learning: Selecting the Right Approach for Real-World Problems

Akash Singh

Indiana Wesleyan University

AIML-500: AI Lab

October 29, 2026

Machine Learning Versus Deep Learning: Selecting the Right Approach for Real-World Problems

Choosing between traditional machine learning (ML) and deep learning (DL) is one of the first decisions in any AI project. The lesson presented traditional ML examples, such as email spam detection, house price prediction, and customer churn prediction, that rely on predefined, human-engineered features. It also presented DL examples, such as image recognition, natural language processing, and autonomous driving, in which neural networks learn features directly from raw data. This report selects one example from each group: customer churn prediction using a Support Vector Machine (SVM) and image recognition using a Convolutional Neural Network (CNN). For each, it identifies a real-world application, explains why the chosen approach fits the problem, and explains why the opposite approach does not.

Example 1: Customer Churn Prediction Using a Support Vector Machine

Real-World Application

Telecommunications providers lose significant revenue when subscribers cancel their service, so many carriers score customers regularly to estimate who is likely to leave. Customers flagged as high risk receive targeted retention offers, such as discounts or plan upgrades, before they cancel. Vafeiadis et al. (2015) compared several machine learning classifiers on telecom churn data and found that SVMs were among the strongest performers, particularly when combined with boosting. This makes churn prediction a well-established, practical use of traditional supervised learning.

Why Traditional Machine Learning Is Suitable

First, churn data is structured and tabular. Each customer is described by meaningful features that the business already understands, such as contract length, tenure, monthly charges, and number of support calls. The predictive signal lives in these features, so there is no need for a model to discover them from raw data.

Second, an SVM is well matched to this kind of binary classification. It finds the boundary that best separates customers who leave from those who stay, and kernel functions allow it to capture nonlinear

relationships between features (Cortes & Vapnik, 1995). It performs well on moderately sized datasets, which is typical of customer records.

Third, the business must act on the results. Retention teams need to understand which factors drive risk, such as month-to-month contracts or rising charges, in order to design an offer. Traditional models are easier to explain, validate, and retrain each month at low computational cost.

Why Deep Learning Is Not Suitable

Deep learning's main advantage is automatically learning features from raw, unstructured inputs such as images, audio, or text. Churn data offers no such advantage because its features are already explicit. Research comparing the two approaches on tabular data found that traditional models, specifically tree-based methods, remained state-of-the-art on medium-sized datasets and trained faster than deep networks (Grinsztajn et al., 2022). A deep network would also require more data, more computing power, and more hyperparameter tuning, and its "black box" nature would make it harder for business leaders to trust and act on its predictions. The added cost and complexity would not produce a meaningful benefit.

Example 2: Image Recognition Using a Convolutional Neural Network

Real-World Application

Apple's Face ID is a widely used facial recognition system. The iPhone's TrueDepth camera projects more than 30,000 infrared dots onto the user's face to build a depth map, and neural networks convert that map and an infrared image into a mathematical representation that is compared to the enrolled face (Apple Inc., n.d.; Verger, 2017). Additional neural networks are trained to detect spoofing attempts using photos or masks, and the system adapts as the user's appearance changes over time (Apple Inc., n.d.). Convolutional architectures are the standard deep learning approach for this kind of visual data (LeCun et al., 2015).

Why Deep Learning Is Suitable

Facial data is high-dimensional and unstructured, consisting of thousands of pixel and depth values rather than a short list of named features. Deep learning excels here because it learns representations in layers: early layers detect edges, middle layers combine them into textures and facial parts, and deeper layers recognize whole faces (LeCun et al., 2015). Because these features are learned from very large training sets, the system can recognize the same person across different angles, lighting conditions, and accessories such as glasses or hats, which Apple states Face ID is designed to handle (Apple Inc., n.d.). Detecting a realistic mask also depends on subtle patterns that no engineer could fully specify by hand, which is exactly the kind of problem deep learning is built to solve.

Why Traditional Machine Learning Is Not Suitable

A traditional algorithm such as an SVM or logistic regression would require engineers to define facial features manually, such as the distance between the eyes or the width of the nose. These hand-crafted

features are brittle: they change with head tilt, shadows, facial hair, or partial coverings, which leads to failed unlocks or false matches. Traditional models also struggle to learn directly from raw pixel data at this scale, so they could not reach the accuracy or security a biometric system requires. For a system protecting personal and financial information, those weaknesses make traditional ML a poor fit.

Comparison Summary

Factor	Customer Churn (Traditional ML)	Face ID (Deep Learning)
Data type	Structured, tabular records	Unstructured images and depth maps
Features	Predefined by humans	Learned automatically by the network
Data volume needed	Moderate	Very large
Computing cost	Low	High
Interpretability	High, supports business action	Low, but accuracy is the priority
Best fit because	Signal is already in known features	Patterns are too complex to define by hand

Conclusion

The choice between traditional machine learning and deep learning should be driven by the nature of the data and the needs of the problem. Customer churn prediction works best with traditional ML because its features are structured, known, and meaningful to the business, and interpretability matters. Facial recognition requires deep learning because its raw visual data contains complex patterns that only a multi-layer network can learn reliably. Matching the method to the problem leads to models that are more accurate, efficient, and useful.

References

Apple Inc. (n.d.). *About Face ID advanced technology*. Apple Support. Retrieved October 29, 2026, from https://support.apple.com/en-us/102381

Cortes, C., & Vapnik, V. (1995). Support-vector networks. *Machine Learning*, *20*(3), 273–297. https://doi.org/10.1007/BF00994018

Grinsztajn, L., Oyallon, E., & Varoquaux, G. (2022). Why do tree-based models still outperform deep learning on typical tabular data? *Advances in Neural Information Processing Systems*, *35*. https://doi.org/10.48550/arXiv.2207.08815

LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *Nature*, *521*(7553), 436–444. https://doi.org/10.1038/nature14539

Vafeiadis, T., Diamantaras, K. I., Sarigiannidis, G., & Chatzisavvas, K. C. (2015). A comparison of machine learning techniques for customer churn prediction. *Simulation Modelling Practice and Theory*, *55*, 1–9. https://doi.org/10.1016/j.simpat.2015.03.003

Verger, R. (2017, September 13). Apple's new Face ID system uses a sensing strategy that dates back decades. *Popular Science*. https://www.popsci.com/apple-face-ID/